

Big Data Analytics Pipeline Using Apache Spark, Hadoop HDFS, Apache Hive and Apache Superset

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1 Introduction

The increasing volume of data generated by e-commerce platforms has made big data technologies essential for data storage, processing, and analysis. Organizations use distributed systems to transform raw transactional data into meaningful business insights that support operational and strategic decision-making.

This project implements an end-to-end big data analytics pipeline using Apache Spark, Hadoop HDFS, Apache Hive, and Apache Superset. The publicly available Olist Brazilian E-Commerce dataset was processed by converting raw CSV files into Parquet format, storing them in Hadoop HDFS, creating Hive external tables, and visualizing the data through an interactive dashboard in Apache Superset.

The main objective of this project is to demonstrate how modern big data tools can be integrated into a complete analytics workflow while providing useful business insights through interactive visualizations.

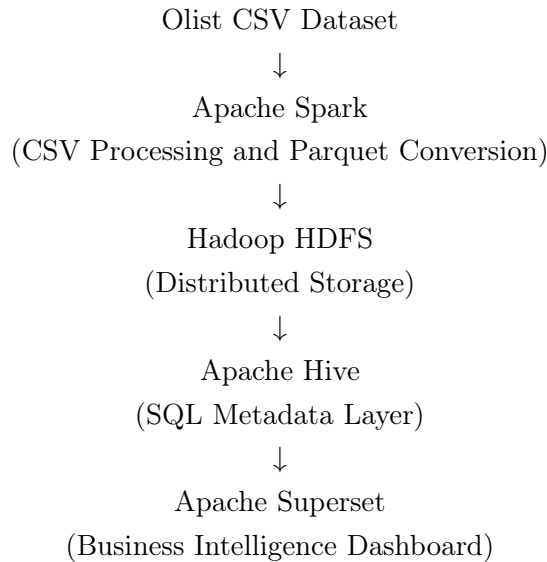
2 Project Objectives

The project was designed with the following objectives:

- Import multiple CSV datasets using Apache Spark.
- Transform raw CSV files into the Parquet format.
- Store processed datasets in Hadoop HDFS.
- Create Hive external tables for SQL querying.
- Connect Apache Superset to Hive.
- Develop an interactive dashboard containing key business metrics.
- Demonstrate an end-to-end big data analytics pipeline.

3 System Architecture

The overall architecture follows a layered big data design where each technology performs a dedicated task within the pipeline.



The pipeline begins with raw CSV files obtained from the Olist dataset. Apache Spark reads every dataset, automatically infers the schema, and converts each table into the Parquet format. Compared to CSV files, Parquet significantly improves analytical query performance because it stores data in a compressed columnar format.

After transformation, the Parquet datasets are uploaded to Hadoop HDFS. HDFS provides reliable distributed storage and allows the datasets to be accessed by multiple services within the analytics environment.

Apache Hive is then configured to create external tables that reference the Parquet files stored inside HDFS. This approach avoids unnecessary data duplication while enabling SQL queries over the distributed storage system.

Finally, Apache Superset connects directly to Hive and provides an interactive interface for dashboard creation and business analysis.

4 Data Processing Pipeline

The original dataset consists of nine CSV files containing information about customers, orders, products, sellers, payments, reviews, and geographical locations.

Apache Spark was selected as the processing engine due to its distributed computing capabilities and efficient support for large-scale data transformation. During the ingestion phase, Spark automatically inferred the schema of each dataset and loaded the CSV files into DataFrames.

Once the datasets were successfully loaded, every table was converted into the Apache Parquet format. This conversion provides several important advantages:

- Faster analytical queries

- Reduced storage requirements through compression
- Column-oriented storage
- Better compatibility with distributed query engines

After the transformation stage, the generated Parquet files were stored inside Hadoop HDFS. Hive external tables were then created on top of these files, allowing SQL-based access without copying the underlying data.

This architecture separates computation, storage, and visualization into independent layers, making the pipeline scalable and easier to maintain.

5 Dashboard Analysis

After completing the data processing pipeline, Apache Superset was used to create an interactive dashboard that summarizes the most important business metrics. The dashboard provides insights into customer behavior, sales performance, seller activity, delivery efficiency, and regional demand. These visualizations allow users to quickly understand key trends and support data-driven decision-making.

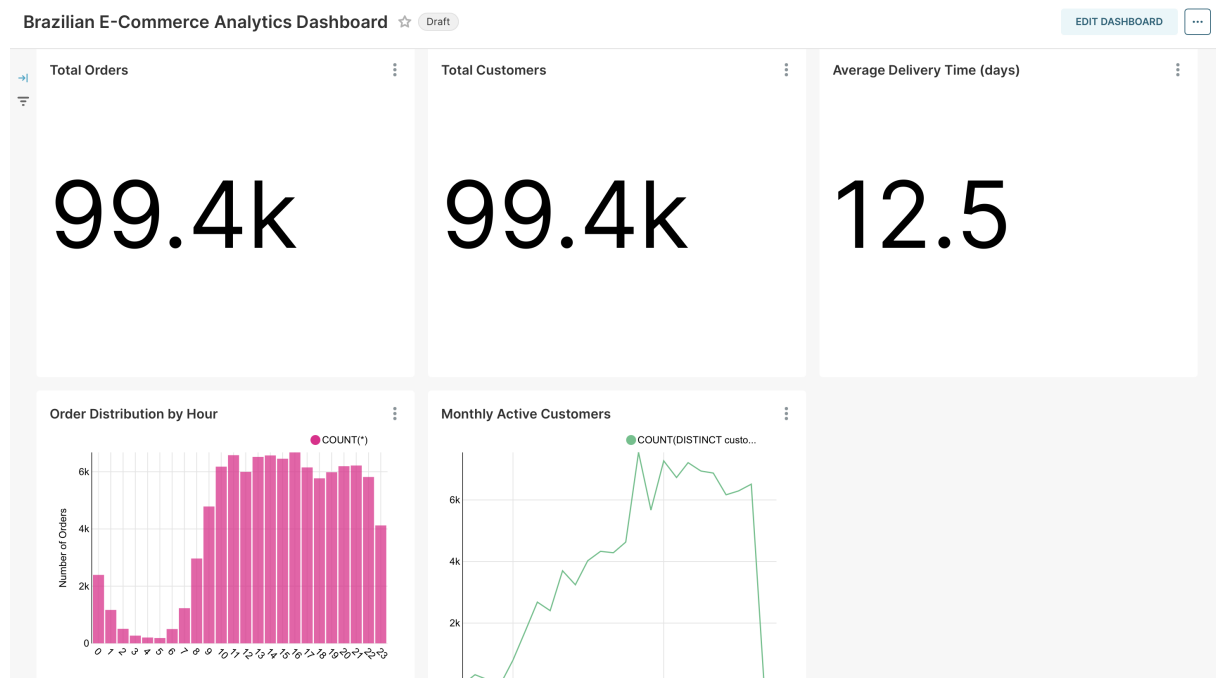
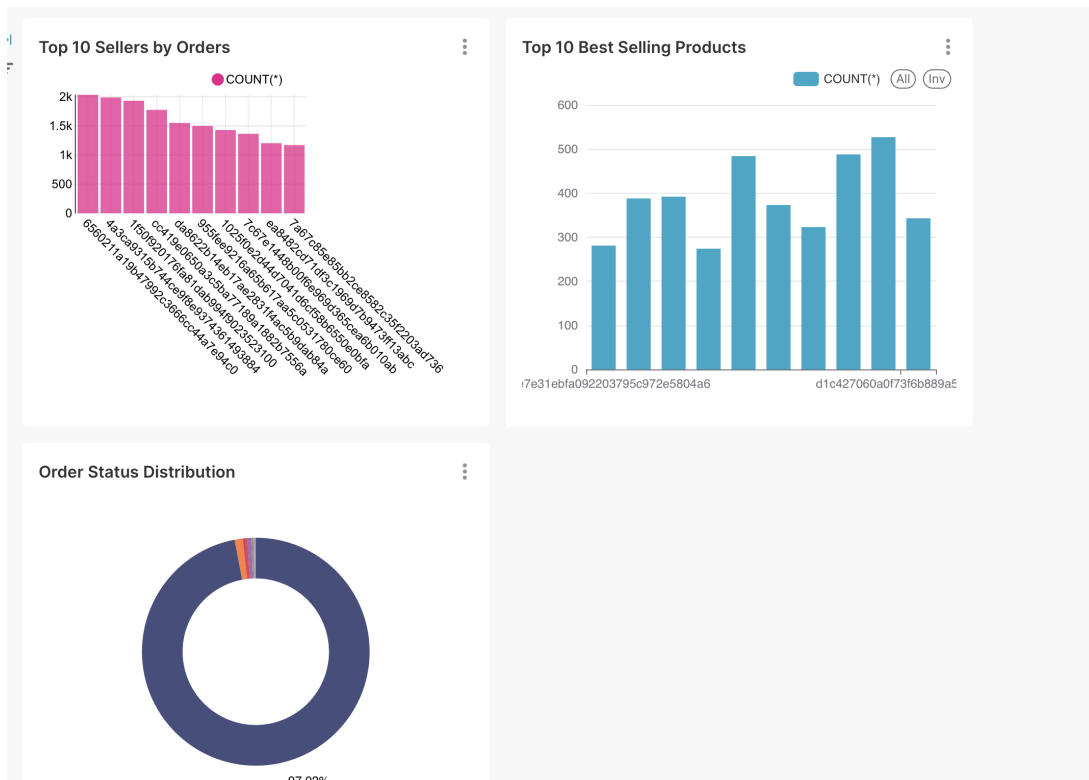
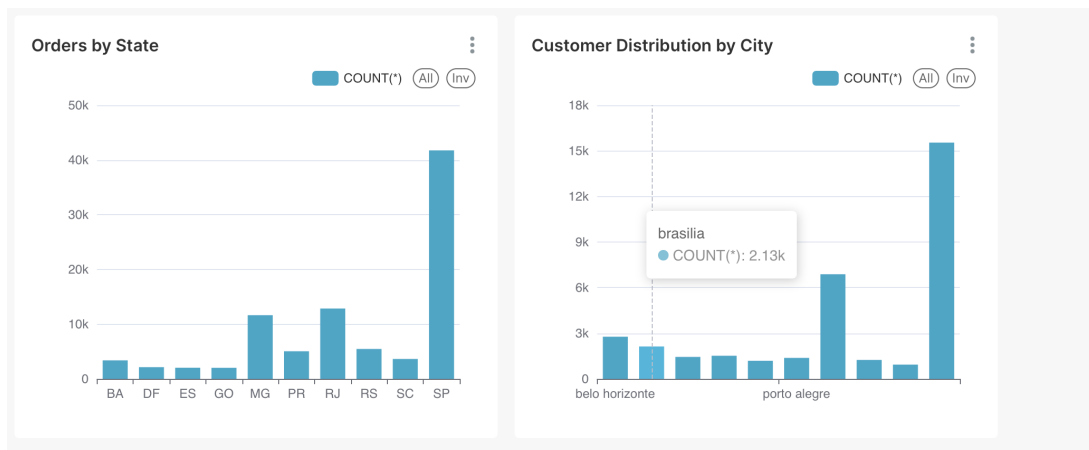


Figure 1: Overall Apache Superset dashboard showing key business metrics.



5.1 Total Orders

The Total Orders KPI displays the overall number of customer orders contained in the dataset. This metric represents the overall business volume of the marketplace and serves as a reference point for evaluating all other business indicators. Monitoring total orders over time enables companies to measure growth, compare different reporting periods, and estimate future demand.

5.2 Total Customers

The Total Customers KPI measures the number of unique customers who placed orders during the observation period. Customer acquisition is one of the most important indicators for e-commerce businesses because sustainable growth depends not only on increasing sales but also on expanding the customer base.

Comparing the number of customers with the total number of orders also provides an indication of customer purchasing behavior.

5.3 Average Delivery Time

Delivery performance is one of the most critical factors affecting customer satisfaction in online shopping.

This visualization calculates the average delivery duration between purchase and successful delivery. Companies continuously monitor this metric to evaluate logistics performance, identify delivery delays, and improve warehouse operations.

Lower delivery times generally contribute to higher customer satisfaction and stronger marketplace performance.

5.4 Monthly Active Customers

The Monthly Active Customers line chart illustrates how customer activity changes over time.

Instead of examining a single summary value, this visualization allows analysts to identify seasonal trends, periods of rapid growth, and fluctuations in customer engagement.

Understanding customer activity patterns supports marketing campaigns, demand forecasting, and long-term business planning.

5.5 Order Status Distribution

This chart presents the distribution of order statuses within the marketplace.

Most successful e-commerce platforms aim for a high percentage of delivered orders while minimizing canceled or unavailable orders.

Monitoring order status distribution enables managers to detect operational issues, evaluate fulfillment performance, and improve customer service processes.

5.6 Orders by State

Customer demand is not distributed equally across different regions.

The Orders by State visualization groups all orders according to Brazilian states, allowing businesses to identify regions with the highest sales activity.

Such geographic analysis supports warehouse planning, regional marketing campaigns, inventory allocation, and logistics optimization.

5.7 Customer Distribution by City

Unlike state-level analysis, this chart focuses on individual cities.

The visualization identifies cities containing the largest customer populations and highlights areas where business demand is concentrated.

Companies frequently use this information when planning regional promotions, delivery centers, and expansion strategies.

5.8 Top 10 Best Selling Products

This visualization ranks the products with the highest number of completed sales.

Best-selling products represent the core revenue generators of an e-commerce platform. Understanding product popularity enables businesses to optimize inventory levels, improve supplier relationships, and prioritize marketing efforts.

The chart also provides valuable information for forecasting future demand.

5.9 Top 10 Sellers by Orders

Marketplace platforms depend heavily on seller performance.

This visualization ranks sellers according to the number of orders they successfully fulfilled.

High-performing sellers contribute significantly to marketplace activity, while low-performing sellers may require operational support or performance evaluation.

Such analysis assists marketplace administrators in monitoring seller productivity.

5.10 Order Distribution by Hour

Customer purchasing behavior varies considerably throughout the day.

This visualization groups orders according to purchase hour, making it possible to identify peak shopping periods.

Knowing when customers are most active allows businesses to optimize advertising schedules, customer support availability, website infrastructure, and promotional campaigns.

Overall, the dashboard transforms transactional data into actionable business information that can support both operational monitoring and strategic decision-making.

6 Discussion

The implemented analytics pipeline demonstrates how multiple open-source technologies can be integrated into a complete business intelligence workflow.

Apache Spark efficiently processed multiple large datasets and transformed them into the Parquet format, significantly improving storage efficiency and analytical performance.

Hadoop HDFS provided a scalable distributed storage environment, while Hive simplified SQL-based data access without duplicating the stored files.

Apache Superset successfully converted the processed datasets into interactive visualizations, allowing business users to explore operational metrics without writing SQL queries.

Although the project focuses primarily on descriptive analytics, the implemented architecture establishes a solid foundation for future machine learning applications, predictive analytics, recommendation systems, and customer segmentation models.

7 Conclusion

This project successfully implemented a complete end-to-end big data analytics pipeline using Apache Spark, Hadoop HDFS, Apache Hive, and Apache Superset.

Starting from raw CSV datasets, the pipeline transformed the data into an optimized analytical format, stored it in a distributed file system, exposed it through Hive, and finally presented business insights through an interactive dashboard.

The developed dashboard summarizes key business indicators including customer growth, order performance, seller activity, geographic distribution, delivery efficiency, and sales trends.

Beyond demonstrating technical implementation, the project illustrates how modern data engineering technologies can support business decision-making by transforming raw data into meaningful analytical information.

Future improvements may include data warehouse modeling, ETL automation, machine learning algorithms, customer segmentation, sales forecasting, and real-time streaming analytics.

References

- [1] Apache Spark. <https://spark.apache.org>
- [2] Apache Hadoop. <https://hadoop.apache.org>
- [3] Apache Hive. <https://hive.apache.org>
- [4] Apache Superset. <https://superset.apache.org>
- [5] Olist Brazilian E-Commerce Dataset. <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>